

Part 1 of 2

Digital Forensics and Investigation



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In a world of cyber crime, an attack is any action that compromises the security policies of the system, carried out through a progression of steps, analogous to the steps of physical attacks. Today, computer hackers are aware of new vulnerabilities because it takes time for organisations to set up adequate system protection. An understanding of hackers and their attack methods and tools used by them is a prerequisite to digital forensics, which is concerned with the acquiring and analysis of evidence of cyber crimes.

What is digital forensics

Digital Crime and Forensic Science in Cyberspace defines digital forensics as the science of collecting evidence often used in a court of law to prosecute those who engage in digital activities that are deemed unlawful. According to *Wikipedia*, "Digital forensics is a branch of forensic science encompassing the recovery and investigation of material found in digital devices, often in relation to the computer crime." A computer crime is defined as any unauthorised use of computer and

data or information without the knowledge of the owner of the computer and data or information. Today, digital forensics is used for investigating the cause to produce evidences, which helps law enforcement agencies to fight against such crimes.

What is digital evidence

Digital evidence is information stored or transmitted in binary form that may be relied upon in court. As per Crime Scene Investigation unit, digital evidence can be found on a computer hard drive, pen drive, mobile phone, personal digital assistant (PDA), compact disk and memory card in a digital camera.

Digital evidence in the form of information is commonly associated with e-crimes such as credit card fraud or child pornography. However, it is now used to prosecute all types of crimes, not just e-crimes. Efforts are made to gather all evidences in the form of digital data, and using computer resources to collect, store and analyse such data is known as computer forensics.

Today, law enforcement agencies need well-trained professionals to collect digital evidence and keep up with rapidly-evolving technologies such as computer operating systems (OSes). Computer crimes are increasing everyday and various acts and methods are being implemented to counter these crimes.

Steps involved in digital forensics and investigation

There are many steps involved in a digital forensic investigation for the collection and analysis of digital evidence. These can be divided into three steps: acquisition, analysis and reporting, as described below.

Acquisition. The process of gathering or collecting electronic evidence is called acquisition or disk imaging or acquiring evidence. It involves creating an exact sector-level duplicate, or forensic duplicate, of the media device. The device that is used

Fig. 1: Various forms of digital evidence (Image courtesy: theregister.co.uk)

